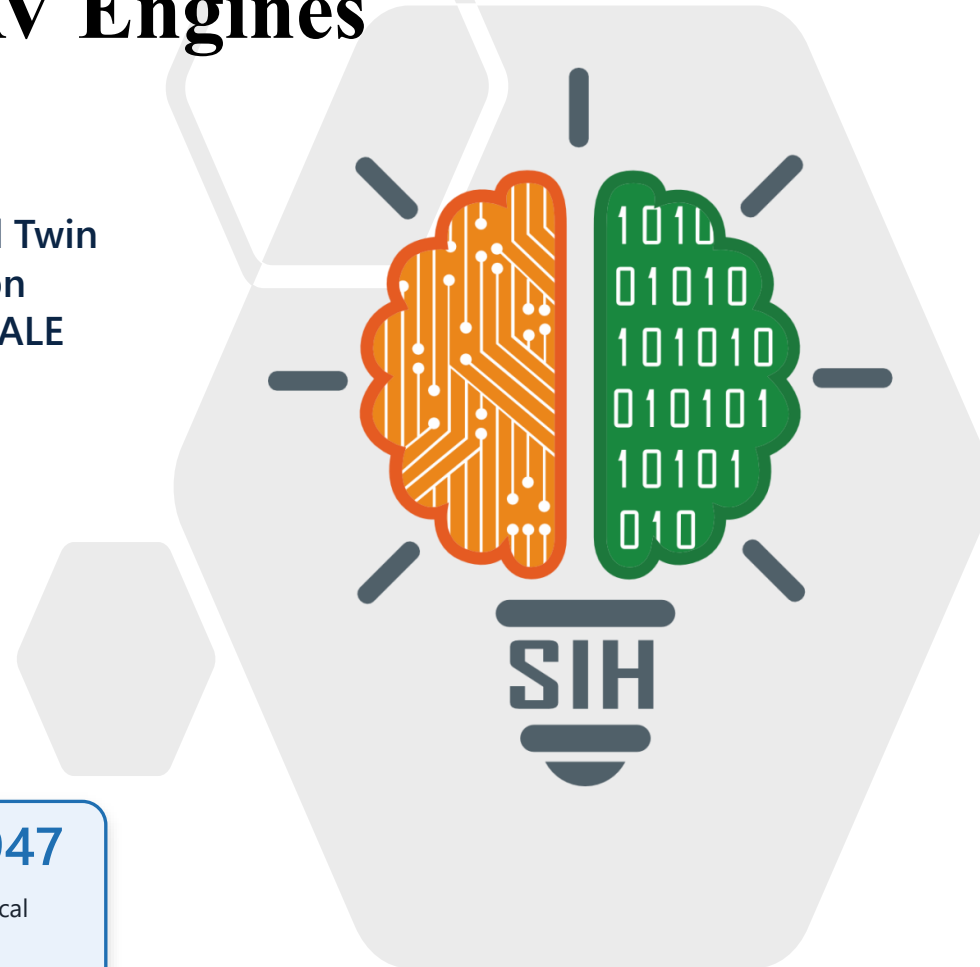


SMART INDIA HACKATHON 2026

GarudaTwin: Propulsion Intelligence for MALE UAV Engines



- Problem Statement ID – SIH26054
- Problem Statement Title – AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs
- Theme – Robotics and Drones
- PS Category – Software
- Team ID – 155188
- Team Name (Registered on portal) – GarudaAstra



 **1.5 s**

AI names an injected engine fault (demo run)

 **0**

false alarms on 15,225 healthy test samples

 **947 / 947**

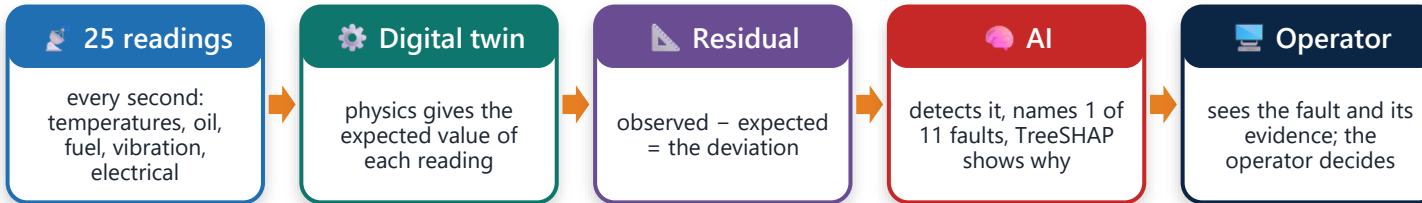
AI decisions identical on replay

Working prototype · every engine reading shown is simulated · measured on held-out simulated test data

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

One idea: a physics digital twin tells the AI what every engine reading *should* be.

Detailed explanation of the proposed solution

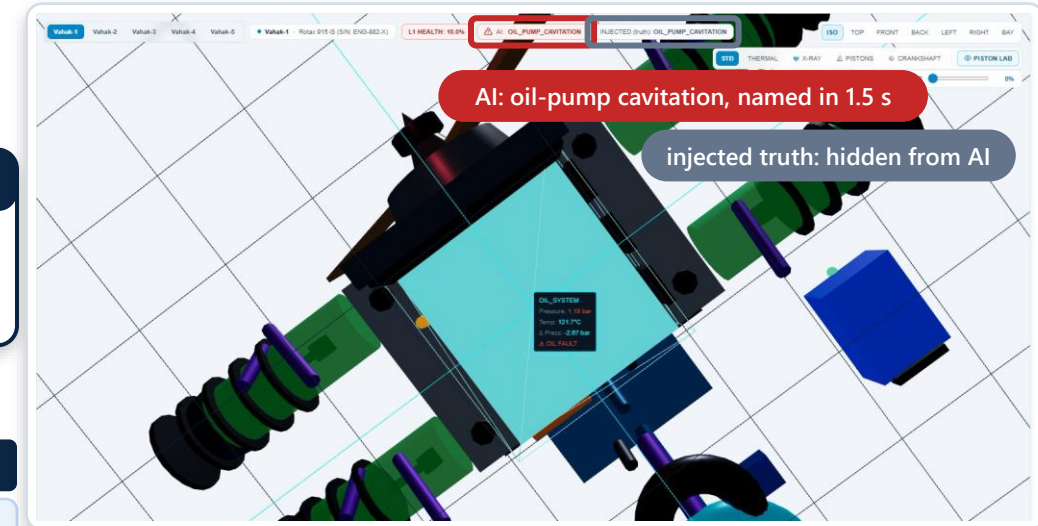


How it addresses the problem

The problem asks for	GarudaTwin's answer	Measured proof (simulated test data)
 Health monitoring	Every reading checked against its expected value, every second	25 readings · lost data → NO DATA in 2 s
 Fault prediction	Catches faults while still small, and names them	96 % of small faults (tuned limits: 37 %) · 11 faults , F1 0.98
 Mission reliability	No false alarms; a lying sensor is not an engine alarm	0 false alarms / 15,225 healthy samples · drifting sensor → engine 99/100

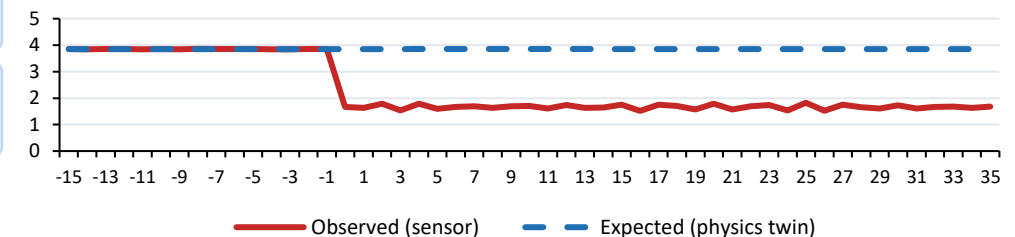
Innovation and uniqueness of the solution

 Physics + AI every alert traces back to a measured deviation	 Sensor or engine? CHT-2 +24 °C, EGT-2 within 1 °C → sensor drift	 Real-data ready CAN bus (DBC) and CSV through the same path	 Checkable truth hidden from the AI; replay 947/947 identical
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Working prototype: 3D engine twin during an injected oil-pump fault (simulated)

Oil pressure (bar): expected vs observed · s from fault onset



Twin expects 3.85 bar, sensor reads 1.67 bar → residual -2.2 bar. Most consistent with oil-pump cavitation; a sensor fault or model error is also possible, so the evidence stays on screen.

Technologies to be used (e.g. programming languages, frameworks, hardware)

Ground-station UI

React Vite Three.js Tailwind CSS Leaflet

Gateway & recorder

Node.js Express Socket.IO SQLite

AI service

Python FastAPI XGBoost scikit-learn
NumPy pandas

Engine bus

CAN bus DBC · cantools python-can

Hardware

Laptop (today) USB-CAN adapter (next)

Methodology and process for implementation (Flow Charts/Images/ working prototype)

① Engine data

Simulator · CAN bus (DBC) · recorded CSV: one path, 25 channels.
CAN virtual rig: 67/67 frames accepted.

✓ DEMONSTRATED

② Gateway

Range & time checks; 20 Hz to the screen.
No frame for 2 s → NO DATA, never NOMINAL.

✓ DEMONSTRATED

③ Golden twin

Physics model with thermal lag.
Throttle, RPM, altitude, air temp → expected value of every reading.

✓ DEMONSTRATED

⑥ Operator GCS

3D twin, evidence table, fleet, what-if, RTB planner.
Advice only: the operator decides.

✓ DEMONSTRATED

⑤ Explainable AI

Mahalanobis anomaly detector (Ledoit-Wolf).
XGBoost names 1 of 11 faults; TreeSHAP shows why.

✓ DEMONSTRATED

④ Residuals

observed – expected, every reading.
→ 45 features incl. 8-sample means.

✓ DEMONSTRATED

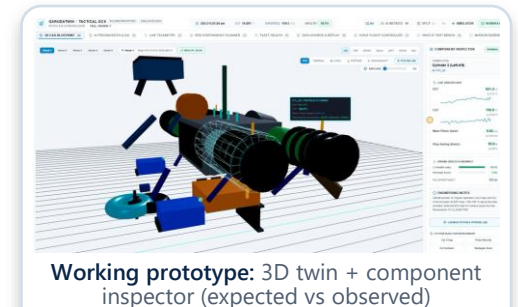
Flight recorder & replay

Every flight saved (SQLite) with the samples the AI scored; any flight or CSV can be replayed.

947 / 947 decisions identical on replay

AI training & testing

378,618 simulated samples · 5,400 episodes · 12 classes
Split by episode 70/15/15: scored only on episodes never seen in training
Macro F1 0.98 (fault naming)



Analysis of the feasibility of the idea

✓ Runs today

Full loop on one laptop:
data → twin → AI → screen

7 test suites

automated tests pass on
a clean install

378,618

simulated training samples,
5,400 episodes

67 / 67

CAN frames accepted from
the virtual rig (DBC decode)

💰 **Economic:** open-source stack, no licence cost; runs on a laptop; a USB-CAN adapter is the only new hardware.

Readiness ladder

Simulation ✓ done

Live CAN path (virtual rig) ✓ done

Engine test bench → next

Flight logs → next

Fleet trial → later

Potential challenges and risks

⚠ **No real engine fault data yet** – all results are on simulated data

⚠ **Uncalibrated sensors** – stress test: fault-naming F1 falls 0.98 → 0.50

⚠ **Twin constants are engineering assumptions** – public specs, not yet validated

⚠ **Operator trust in an AI** – a wrong call must never act on its own

⚠ **Remaining life needs failure-progression data**

Strategies for overcoming these challenges

✓ Engine test-bench and flight logs; calibrate the twin per engine; repeat the same comparison on real data

✓ Calibrate on each engine's healthy runs; sensor cross-checks; train with offset variety

✓ Fit the constants from test-bench runs; keep data-quality and NO DATA checks

✓ Evidence on screen, truth hidden from the AI, operator decides; nothing automatic

✓ Kept as research until run-to-failure or maintenance records exist

IMPACT AND BENEFITS

 **Measured: same 810 held-out simulated test episodes for every method**

48,920 test samples (15,225 healthy) · an alarm needs 2 consecutive samples · simulator results, not yet real-engine results

Method	Faults caught	Small faults (5–15 %)	False alarms	Names it
Fixed limits (hand-set)	26.3 %	20 %	2,110	No
Limits tuned on healthy data	68.3 %	37 %	137	No
+ physics twin	93.1 %	90 %	5	No
+ AI = GarudaTwin	96.5 %	96 %	0	Yes (F1 0.98)

FALSE ALARMS ON HEALTHY ENGINES

2,110 → 0

SMALL FAULTS CAUGHT

37 % → 96 %

tuned fixed limits → physics twin + AI, and it names the fault

Potential impact on the target audience



UAV operators: an earlier, explained warning; a lying sensor is not an engine alarm; lost data is never shown as healthy



Maintenance teams: the residual pattern shows where to look first; every flight can be replayed and checked



Mission planners: fewer missions lost to false alarms; route options stay advice, the operator decides

Benefits of the solution (social, economic, environmental, etc.)



Social / safety

fewer engine-related aircraft losses; explained alerts operators can trust



Economic

maintain on condition; avoid needless part removals and aborted sorties



Environmental

fewer wasted parts, less fuel burnt on repeat test flights



Strategic

open, explainable engine-health tool built in India for Indian UAV fleets

Expected benefits; to be confirmed on real engine data.

Details / Links of the reference and research work

13 verified sources, *mapped onto the six ISO 13374 condition-monitoring blocks*

DA Engine data

ENGINE-CLASS BASELINE (MALE UAV)
Nugroho, R. C. et al. (2023)
 Performance simulation of Rotax 915iS PUNA MALE engine using AVL Boost
AIP Conf. Proc. 2981, 020013
[doi:10.1063/5.0184501](https://doi.org/10.1063/5.0184501)

CAN BUS: OUR LIVE INPUT
ISO 11898-1:2024
 Road vehicles — Controller area network (CAN) — Part 1: Data link layer
International standard (current edition)
iso.org/standard/86384

DM Golden twin

THE DIGITAL-TWIN CONCEPT
Glaessgen, E. & Stargel, D. (2012)
 The Digital Twin Paradigm for Future NASA and U.S. Air Force Vehicles
53rd AIAA SDM Conference
[doi:10.2514/6.2012-1818](https://doi.org/10.2514/6.2012-1818)

ONLINE TWIN ON A UAV ENGINE
Peng, C.-C. & Chen, Y.-H. (2024)
 Fixed-Wing UAV Rotary Engine Anomaly Detection via Online Digital Twin Methods
IEEE Trans. Aerosp. Electron. Syst. 60(1)
[doi:10.1109/TAES.2023.3329797](https://doi.org/10.1109/TAES.2023.3329797)

SD Residual check

RESIDUAL = OBSERVED – MODEL
Isermann, R. (2005)
 Model-based fault-detection and diagnosis – status and applications
Annual Reviews in Control 29(1), 71–85
[doi:10.1016/j.arcontrol.2004.12.002](https://doi.org/10.1016/j.arcontrol.2004.12.002)

ANOMALY SCORE · INDIAN SCIENCE
Mahalanobis, P. C. (1936)
 On the generalised distance in statistics
Reprinted in Sankhyā A 80(S1), 2018
[doi:10.1007/s13171-019-00164-5](https://doi.org/10.1007/s13171-019-00164-5)
 + Ledoit & Wolf (2004) covariance

HA Fault diagnosis

FAULT CLASSIFIER
Chen, T. & Guestrin, C. (2016)
 XGBoost: A Scalable Tree Boosting System
ACM SIGKDD (KDD '16), 785–794
[doi:10.1145/2939672.2939785](https://doi.org/10.1145/2939672.2939785)

PISTON-ENGINE FAULT SIGNS
Miljković, D. (2019)
 Detecting Aircraft Piston Engine Problems by Analysis of Engine Parameters
MIPRO 2019, 937–942
[doi:10.23919/MIPRO.2019.8757079](https://doi.org/10.23919/MIPRO.2019.8757079)

PA Remaining life

SIMULATED RUN-TO-FAILURE (NASA)
Saxena, A. et al. (2008)
 Damage propagation modeling for aircraft engine run-to-failure simulation
Int. Conf. Prognostics & Health Mgmt. (PHM)
[doi:10.1109/PHM.2008.4711414](https://doi.org/10.1109/PHM.2008.4711414)

 **OUR STATUS**
Research in progress, not claimed.
 Remaining life needs real failure-progression data; the twin and AI above do not depend on it.

AG Explain & advise

TreeSHAP: WHY THE AI DECIDED
Lundberg, S. M. et al. (2020)
 From local explanations to global understanding with explainable AI for trees
Nature Machine Intelligence 2(1), 56–67
[doi:10.1038/s42256-019-0138-9](https://doi.org/10.1038/s42256-019-0138-9)

THE 6-BLOCK CHAIN ABOVE
ISO 13374-1:2003
 Condition monitoring and diagnostics of machines — Data processing, communication and presentation
International standard, Part 1
iso.org/standard/21832

 Engine facts: BRP-Rotax official data

Rotax 915 iS: 1,352 cm³ · 104 kW (141 hp) take-off · 2.54 reduction gearbox · max operating altitude 23,000 ft

→ **Our dataset spans 0–23,000 ft: the engine's whole published altitude range.**

flyrotax.com/products/915-is-a-isc-a

Twin constants beyond these specs are documented engineering assumptions; calibrating them on a real engine is our next step.

 What GarudaTwin adds to this research

-  Names the fault, shows its evidence (piston engine, 11 faults)
-  Lying sensor ≠ sick engine (sensor-vs-engine check)
-  Silence is never healthy (NO DATA within 2 s)
-  False alarms 2,110 → 0 vs simple limits (simulated test data)

 Code


GitHub: code, tests, model card
github.com/suhruthsai/GarudaTwin-Propulsion-Intelligence

 Datasets


Google Drive: all data + dataset card
drive.google.com/drive/folders/1NzgCeVztNYcLeIkmXkrBK1E8dtII7Up